CampusNotes - Academic Study Resource

DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING - SEMESTER 5

OFFICIAL ACADEMIC STUDY MATERIAL - CSE CURRICULUM

Assignment 1: Asymptotic Notations & Recurrence Relations

Course: Design Analysis and Algorithms [23CS3502]
Category: Assignments | Assignment #1 | Academic Year: 2026-27

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1. Overview & Objective:

Formal mathematical analysis of time and space complexity bounds.

2. Assignment Problems & Instructions:

1. Prove formally that 3n^2 + 5n log n is O n^2 using asymptotic definitions.
2. Solve recurrence T n = 4T n/2 + n^2 using the Master Theorem.
3. Solve T n = 2T n/2 + n log n using Recursion Tree and Substitution methods.
4. Analyze worst-case, best-case, and average-case bounds of binary search.

Submission Guidelines & Verification:

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